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Ventilation apparatus

Abstract

A ventilation apparatus includes a housing having an outdoor air inlet, an outdoor air outlet, an indoor air inlet, and an indoor air outlet; a partition dividing an inner space of the housing into a ventilation space and an air conditioning space; a total heat exchange element inside the ventilation space and having four side edges respectively disposed to face four side surfaces of the housing; four partition walls configured to respectively connect the four side edges of the total heat exchange element and the four side surfaces of the housing to partition the ventilation space into an outdoor air discharge space, an indoor air inlet space, and an indoor air discharge space; guide ducts respectively provided in the outdoor air discharge space, the indoor air inlet space, and the indoor air discharge space; and a bypass duct branching from the guide duct provided in the indoor air inlet space.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) The present application claims the benefits of priority to Korean Patent Application No. 10-2022-0095434, filed on Aug. 1, 2022, of which is incorporated herein by reference in its entirety.

BACKGROUND

1. Field

(2) The present disclosure relates to a ventilation apparatus.

2. Description of the Related Art

(3) A ventilation apparatus is an apparatus for discharging indoor air to the outside and supplying fresh outdoor air to the indoor space, and the main component of the ventilation apparatus is a total heat exchange element that allows only heat exchange without mixing the discharged indoor air and the introduced outdoor air.

(4) Recently, a complex ventilation apparatus capable of performing a cooling function in addition to a ventilation function through total heat exchange has emerged.

(5) Korean Patent Registration No. 10-2124364 (Jun. 18, 2020) discloses a complex ventilation apparatus in which a refrigeration system that forms a cooling cycle using a refrigerant inside the ventilation apparatus is mounted.

(6) In the ventilation apparatus disclosed in the prior art, the components for cooling and heating, that is, a compressor, a condenser (second heat exchanger), and an evaporator (first heat exchanger) are installed inside the ventilation apparatus, and a damper (defined as a second damper) is provided for rapid cooling in which discharged indoor air passes through the evaporator rather than the total heat exchange element and then flows back into the indoor space.

(7) The conventional complex ventilation apparatus having the above-described structure has the following disadvantages.

(8) First, since both the evaporator and the compressor are provided inside the ventilation apparatus, there is a disadvantage in that a supply fan and an exhaust fan of the ventilation apparatus have to be driven simultaneously in an operation mode in which a cooling cycle is driven. For example, when the indoor rapid cooling operation is performed by closing the first damper and opening the second damper while the cooling cycle is driven, the exhaust fan has to operate for cooling the condenser. In the indoor rapid cooling operation mode, the outdoor air introduced by the operation of the exhaust fan has to be bypassed without passing through the total heat exchange element, so that heat is absorbed from the condenser and then discharged back to the outside through an exhaust discharge passage. To this end, a separate damper has to be provided to guide the introduced outdoor air to the exhaust discharge passage. As described above, when the compressor and the condenser are installed inside the ventilation apparatus, there is a disadvantage in that at least four dampers have to be provided.

(9) Therefore, since the exhaust fan has to necessarily operate to cool the condenser in the rapid cooling mode, there is a disadvantage in that power consumption increases.

(10) Second, since the heavy evaporator and compressor are all provided inside the ventilation apparatus, the volume and weight of the ventilation apparatus increase. Accordingly, there is a difficulty in installing the ventilation apparatus, and there is a risk of falling due to gravity after installation.

(11) Third, if there is a problem in the operation of the compressor, there is a disadvantage in that the repair is not easy.

(12) Fourth, since a passage for discharging contaminated indoor air directly to the outside without passing through the total heat exchange element is not provided inside the ventilation apparatus, there is a disadvantage in that the life of the total heat exchange element is shortened. For example,

if fish or meat is cooked, indoor air containing odors and smoke passes through the total heat exchange element, and thus, animal fat with high viscosity is adsorbed on the surface of the total heat exchange element. This adsorption phenomenon increases the flow resistance of the exhausted indoor air and acts as a factor of lowering the total heat exchange efficiency with the introduced outdoor air.

(13) Fifth, since the indoor air discharged to the outside has to pass through the total heat exchange element, the outdoor air passing through the total heat exchange element absorbs heat from the discharged indoor air when the indoor temperature is higher than the outdoor temperature. As a result, since fresh air with a relatively low temperature cannot be supplied to the indoor space, there is a disadvantage in that a user does not feel a ventilation effect in summer.

(14) Sixth, in the prior art, since air is purified while the introduced outdoor air passes through the total heat exchange element and then passes through various filters, various contaminants including dust may be accumulated inside the total heat exchange element, and harmful bacteria may proliferate while being attached to the surface of the total heat exchange element.

(15) Seventh, since it is not possible to install only a ventilation apparatus provided with a ventilation system or selectively combine an air conditioner provided with a cooling system in a state where a ventilation apparatus provided with a ventilation system is purchased, there is a disadvantage in that the range of product selection of consumers is narrow.

(16) Eighth, in the case of the prior art, since it can be designed only for cooling as an integrated outdoor unit, there is a disadvantage in that it is difficult to implement a heating operation.

SUMMARY

(17) The present disclosure is proposed to solve the above problems.

(18) To achieve the above objects, a ventilation apparatus includes: a housing provided with a flange connector including an outdoor air inlet, an outdoor air outlet, an indoor air inlet, and an indoor air outlet; a partition configured to partition an inner space of the housing into a ventilation unit space at an upper side and an air conditioning unit space at a lower side; a total heat exchange element erected in a hexahedral shape inside the ventilation unit space and having four side edges respectively disposed to face four side surfaces of the housing; four partition walls configured to respectively connect the four side edges of the total heat exchange element and the four side surfaces of the housing to partition the ventilation unit space except a space for the total heat exchange element into an outdoor air inlet space, an outdoor air discharge space, an indoor air inlet space, and an indoor air discharge space; guide ducts respectively erected in the outdoor air discharge space, the indoor air inlet space, and the indoor air discharge space; a suction fan module disposed in the outdoor air discharge space and configured to supply outdoor air sucked through the outdoor air inlet into an indoor space; an exhaust fan module disposed in the indoor air discharge space and configured to discharge indoor air sucked through the indoor air inlet to an outside; and a bypass duct branching from a guide duct erected in the indoor air inlet space, bypassing the total heat exchange element, and extending to the indoor air discharge space.

(19) The guide duct includes: an outdoor air discharge guide duct erected in the outdoor air discharge space; an indoor air inlet guide duct erected in the indoor air inlet space; and an indoor air discharge guide duct erected in the indoor air discharge space, wherein the bypass duct is branched from a side surface of the indoor air inlet guide duct.

(20) The bypass duct is bent along an upper surface of the partition and an inner edge of the housing and communicates with the indoor air outlet.

(21) The ventilation apparatus further includes a bypass damper module mounted at an inlet of the bypass duct.

(22) The outdoor air inlet communicates with the outdoor air inlet space, the outdoor air outlet communicates with an upper surface of the outdoor air discharge guide duct, the indoor air inlet communicates with an upper surface of the indoor air inlet guide duct, and the indoor air outlet communicates with an upper surface of the indoor air discharge guide duct.

- (23) The ventilation apparatus further includes: a duct flange mounted on an upper surface of the housing and fixed to a position corresponding to the flange connector; and an air duct connected to the duct flange and extending upward from the housing.
- (24) The duct flange includes: an outdoor air inlet flange mounted at the outdoor air inlet; an outdoor air discharge flange mounted at the outdoor air outlet; an indoor air inlet flange mounted at the indoor air inlet; and an indoor air discharge flange mounted at the indoor air outlet.
- (25) The air duct includes: an outdoor air inlet duct coupled to the outdoor air inlet flange; an outdoor air discharge duct coupled to the outdoor air discharge flange; an indoor air inlet duct coupled to the indoor air inlet flange; and an indoor air discharge duct coupled to the indoor air discharge flange.
- (26) The ventilation apparatus further includes: a boundary wall configured to partition the air conditioning unit space into first and second spaces; a heat exchanger disposed in the first space; and an air conditioning fan module disposed in the second space.
- (27) The partition includes: an air conditioning inlet configured to connect the indoor air inlet guide duct and the first space; and an air conditioning outlet configured to connect the outdoor air discharge guide duct and the second space, and an air conditioning damper module is mounted at each of the air conditioning inlet and the air conditioning outlet.
- (28) A total heat exchange outlet is formed on a side surface of the indoor air inlet guide duct, a total heat exchange damper module is mounted at the total heat exchange outlet, and the total heat exchange damper module operates so that an inner space of the indoor air inlet guide duct and the indoor air inlet space selectively communicate with each other.
- (29) The ventilation apparatus further includes a filter module mounted on a side surface of the total heat exchange element communicating with the outdoor air inlet space and configured to purify introduced outdoor air, wherein the filter module includes at least one of a HEPA filter and a pre-filter.
- (30) The ventilation apparatus having the above-described configuration according to an embodiment of the present disclosure has the following effects.
- (31) First, the ventilation system and the air conditioning system are placed up and down, and the duct connection portion is formed on the upper surface of the ventilation apparatus. Accordingly, there is an advantage in that the ventilation apparatus can be installed in the stand form on the indoor floor. When installed in the stand form, there is an advantage in that the risk of falling is eliminated compared to when installed on the ceiling.
- (32) Second, since the air conditioning area is disposed below the ventilation area, the space of the ventilation area can be sufficiently secured. As a result, it is possible to increase the size of the total heat exchange element, there is an advantage in that ventilation performance is improved.
- (33) Third, since the condenser is not installed in the ventilation area, there is no need to drive the exhaust fan in the quick cooling mode which bypasses the indoor air to pass through the evaporator and then discharges the indoor air to the indoor space. Accordingly, there is an advantage in that power consumption is reduced.
- (34) Fourth, since only one of two heat exchangers constituting the cooling cycle is installed in the air conditioning area, there is an advantage in that the heat exchanger in the air conditioning area operates as an evaporator in the cooling mode through the operation of the four-way valve installed at the outlet of the compressor, and operates as a condenser in the heating mode.
- (35) Fifth, since the bypass passage is provided inside the ventilation apparatus so that the indoor air is discharged directly to the outside without passing through the total heat exchange element, there is an advantage in that the life of the total heat exchange element is prolonged by minimizing the contamination of the total heat exchange element.
- (36) Sixth, since it is possible to discharge the indoor air through the bypass passage, the introduced outdoor air can be supplied to the indoor space without recovering waste heat contained in the indoor air. Accordingly, there is an advantage in that ventilation performance and user

satisfaction are improved.

(37) Seventh, as outdoor air passes through various filters before the outdoor air is introduced into the total heat exchange element, foreign matters and harmful bacteria are filtered out in advance.

Accordingly, there is an advantage in that the life of the total heat exchange element is prolonged.

(38) Eighth, since the ventilation means including the total heat exchange element and the filters are provided in the form of a single module, there is an advantage of easy installation and repair.

(39) Ninth, since the apparatus constituting the air conditioning system is detachably coupled to the lower side of the apparatus constituting the ventilation system in the form of a module, there is an advantage in that the range of product selection of consumers is widened.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is an external perspective view of a ventilation apparatus according to an embodiment of the present disclosure.

(2) FIG. 2 is a perspective view showing the internal structure of the ventilation apparatus when viewed from a first side in a state in which a housing is removed.

(3) FIG. 3 is a perspective view showing the internal structure of the ventilation apparatus when viewed from a second side opposite to the first side in a state in which the housing is removed.

(4) FIG. 4 is a bottom perspective view of the ventilation apparatus when viewed upward at the first side in a state in which the housing is removed.

(5) FIG. 5 is a bottom perspective view of the ventilation apparatus when viewed upward at the second side in a state in which the housing is removed.

(6) FIG. 6 is an exploded perspective view showing a structure of an interface between a ventilation unit and an air conditioning unit.

(7) FIG. 7 is an exploded perspective view showing the internal configuration of the ventilation apparatus according to an embodiment of the present disclosure.

(8) FIG. 8 is a cutaway perspective view of an indoor air inlet guide duct and a bypass duct taken along line 8-8 of FIG. 7.

(9) FIG. 9 is a side view of a guide duct showing an installation structure of a guide duct and a duct flange.

(10) FIG. 10 is a plan view of the guide duct showing the installation structure of the guide duct and the duct flange.

(11) FIG. 11 is a plan view of the duct flange from which the duct flange is removed.

(12) FIG. 12 is a view showing air flow inside the ventilation apparatus in a total heat exchange ventilation mode.

(13) FIG. 13 is a view showing air flow inside the ventilation apparatus in a quick ventilation mode.

(14) FIG. 14 is a view showing air flow inside the ventilation apparatus in a quick cooling mode.

(15) FIG. 15 is a view showing air flow inside the ventilation apparatus in a total heat exchange ventilation/cooling simultaneous mode.

(16) FIG. 16 is a view showing air flow inside the ventilation apparatus in a quick ventilation/cooling simultaneously mode.

(17) FIG. 17 is an exploded perspective view of the ventilation apparatus showing a state in which the ventilation unit and the air conditioning unit are connected by a connection unit.

(18) FIG. 18 is a longitudinal cross-sectional view taken along line 18-18 of FIG. 1.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(19) Hereinafter, a ventilation apparatus according to an embodiment of the present disclosure will be described in detail with reference to the accompanying drawings.

(20) FIG. 1 is an external perspective view of a ventilation apparatus according to an embodiment of the present disclosure, FIG. 2 is a perspective view showing the internal structure of the ventilation apparatus when viewed from a first side in a state in which a housing is removed, FIG. 3 is a perspective view showing the internal structure of the ventilation apparatus when viewed from a second side opposite to the first side in a state in which the housing is removed, FIG. 4 is a bottom perspective view of the ventilation apparatus when viewed upward at the first side in a state in which the housing is removed, and FIG. 5 is a bottom perspective view of the ventilation apparatus when viewed upward at the second side in a state in which the housing is removed.

(21) Referring to FIGS. 1 to 5, a ventilation apparatus **10** according to an embodiment of the present disclosure includes a ventilation unit **10**, an air conditioning unit **30** detachably coupled to the bottom surface of the ventilation unit **10**, and a connection unit **70** connecting the air conditioning unit **30** to the ventilation unit **20**.

(22) The connection unit **70** connects both sides of a hexahedral upper housing **201** constituting the ventilation unit **20** and both sides of a hexahedral lower housing **301** constituting the air conditioning unit **30**.

(23) When the ventilation unit **20** and the air conditioning unit **30** are provided as non-separable types, the ventilation unit **20** and the air conditioning unit **30** may be accommodated in one large housing. Specifically, a partition that partitions the inner space of the housing up and down may be provided inside the housing, components constituting the ventilation unit **20** may be accommodated in the upper side of the partition, and components constituting the air conditioning unit **30** may be accommodated in the lower side of the partition. Communication holes, that is, an air conditioning inlet and an air conditioning outlet, which will be described later, may be formed in the partition, respectively.

(24) In detail, the ventilation apparatus **10** further includes a duct flange **50** coupled to the upper surface of the ventilation unit **10**, and an air duct **40** connected to the duct flange **50**.

(25) A flange connector **2010** is formed on the upper surface of the upper housing **201**. An outdoor air inlet **2011**, an outdoor air outlet **2012**, an indoor air inlet **2013**, and an indoor air outlet **2014** are formed in the flange connector **2010**. The outdoor air inlet **2011** is formed at a position facing the outdoor air outlet **2012** in a diagonal direction, and the indoor air inlet **2013** is formed in a position facing the indoor air outlet **2014** in a diagonal direction.

(26) The duct flange **50** is mounted to the flange connector **2010**. The duct flange **50** includes an outdoor air inlet flange **51** connected to the outdoor air inlet **2011**, an outdoor air discharge flange **52** connected to the outdoor air outlet **2012**, an indoor air inlet flange **53** connected to the indoor air inlet **2013**, and an indoor air discharge flange **54** connected to the indoor air outlet **2014**.

(27) The air duct **40** includes an outdoor air inlet duct **41** connected to the outdoor air inlet flange **51**, an outdoor air discharge duct **42** connected to the outdoor air discharge flange **52**, an indoor air inlet duct **43** connected to the indoor air inlet flange **53**, and an indoor air discharge duct **44** connected to the indoor air discharge flange **54**.

(28) Meanwhile, the ventilation unit **20** further includes a ventilation module **23** disposed in the inner center of the upper housing **201**.

(29) The ventilation module **23** may include a total heat exchange element **231** that only exchanges heat without mixing indoor air and outdoor air, a HEPA filter **232** disposed on one side of the total heat exchange element **231**, and a pre-filter **233** disposed on one side of the HEPA filter **232**.

(30) The total heat exchange element **231** has a hexahedral shape having upper and lower surfaces and four side surfaces. The four side surfaces include two side surfaces through which outdoor air passes and two side surfaces through which indoor air passes. An outdoor air passage and an indoor air passage cross each other inside the total heat exchange element **231**. A passage through which outdoor air flows and a passage through which indoor air flows are alternately arranged in the height direction of the total heat exchange element **231**.

(31) One surface of the HEPA filter **232** is in close contact with the side surface through which

outdoor air is introduced among the four side surfaces of the total heat exchange element **231**, and the pre-filter **233** is in close contact with the other surface of the HEPA filter **232**. Accordingly, outdoor air introduced through the outdoor air inlet duct **41** is purified while sequentially passing through the pre-filter **233** and the HEPA filter **232** and then passes through the total heat exchange element **231**.

(32) The four side surfaces of the total heat exchange element **231** are erected inside the upper housing **201** in a form facing each of the four corners of the upper housing **201**. Four partition walls **28** are disposed between four side edges forming the total heat exchange element and four side surfaces of the upper housing **201**. Accordingly, a space inside the upper housing **201** excluding a space occupied by the total heat exchange element **231** is partitioned into four spaces by the four partition walls **28**.

(33) The four partitioned spaces are defined as an outdoor air inlet space **S1**, an outdoor air discharge space **S2**, an indoor air inlet space **S3**, and an indoor air discharge space **S4**.

(34) The outdoor air inlet **2011** is formed on the upper surface of the outdoor air inlet space **S1**, and the outdoor air outlet **2012** is formed on the upper surface of the outdoor air discharge space **S2**. The indoor air inlet **2013** is formed on the upper surface of the indoor air inlet space **S3**, and the indoor air outlet **2014** is formed on the upper surface of the indoor air discharge space **S4**.

(35) In addition, the ventilation unit **20** includes an indoor air inlet guide duct **24** erected in the indoor air inlet space **S3**, an outdoor air discharge guide duct **25** erected in the outdoor air discharge space **S2**, an indoor air discharge guide duct **26** erected in the indoor air discharge space **S4**, and a bypass duct **27** extending from one side surface of the indoor air inlet guide duct **24** and connected to the indoor air discharge space **S4**.

(36) The bypass duct **27** extends and bends along the edge of the upper housing **201** from one side surface of the indoor air inlet guide duct **24** to bypass the ventilation module **23**, passes through the outdoor air inlet space **S1** and extends to the indoor air discharge space **S4**. That is, the outlet of the bypass duct **27** communicates with the indoor air discharge space **S4**.

(37) In addition, the ventilation unit **20** includes a suction fan module **21** disposed in the outdoor air discharge space **S2** and having an outlet connected to one side surface of the outdoor air discharge guide duct **25**, and an exhaust fan module **22** disposed in the indoor air discharge space **S4** and connected to one side surface of the indoor air discharge guide duct **26**.

(38) The suction fan module **21** includes a fan housing **211**, a suction fan **212** accommodated in the fan housing **211**, and a fan motor **213** configured to drive the suction fan **212**. Like the suction fan module **21**, the exhaust fan module **22** also includes a fan housing **221**, an exhaust fan **222**, and a fan motor **223**.

(39) Meanwhile, the air conditioning unit **30** includes an evaporation fan module **34** accommodated in the lower housing **301**, an evaporator **32**, and a drain pan **33** disposed under the evaporator **32**. Like the suction fan module **21** or the exhaust fan module **22**, the evaporation fan module **34** also includes a fan housing, an evaporation fan, and a fan motor.

(40) When a refrigerant cycle operates as a heat pump, the evaporator **32** functions as a condenser and the evaporation fan module **34** functions as a condensing fan module. Therefore, as a term inclusive of all of these, the evaporator **32** may be defined as a heat exchanger, and the evaporation fan module **34** may be defined as an air conditioning fan module.

(41) A boundary wall **31** may divide the inner space of the lower housing **301** into an evaporator space in which the evaporator **32** is accommodated and an evaporation fan space in which the evaporation fan module **33** is accommodated.

(42) A communication hole **311** is formed in the boundary wall **31**, and the outlet of the evaporation fan module **34** is coupled to the communication hole **311**. Therefore, the air introduced into the evaporation fan space passes through the boundary wall **31** by the evaporation fan module **34** and is guided to the evaporator space. The air guided to the evaporator space exchanges heat with the evaporator **32**.

(43) In addition, an air conditioning inlet **101** and an air conditioning outlet **102** are formed in the interface between the ventilation unit **20** and the air conditioning unit **30** (see FIG. 6). An air conditioning damper module **60** is mounted on the air conditioning inlet **101** and the air conditioning outlet **102**. The air conditioning damper module **60** includes an inlet-side air conditioning damper module **61** configured to open or close the air conditioning inlet **101** and an outlet-side air conditioning damper module **62** configured to open or close the air conditioning outlet **102**.

(44) The air conditioning inlet **101** communicates with the indoor air inlet guide duct **24**, and the air conditioning outlet **102** communicates with the outdoor air discharge guide duct **25**.

(45) FIG. 6 is an exploded perspective view showing the structure of the interface between the ventilation unit and the air conditioning unit.

(46) Referring to FIG. 6, the upper surface **302** of the air conditioning unit **30** is in close contact with the lower surface **202** of the ventilation unit **20**.

(47) In detail, a first opening **2021** forming the air conditioning inlet **101** and a second opening **2022** forming the air conditioning outlet **102** are formed in the lower surface **202** of the ventilation unit **20**.

(48) A first opening **3021** forming the air conditioning inlet **101** and a second opening **3022** forming the air conditioning outlet **102** are formed in the upper surface **302** of the air conditioning unit **30**.

(49) The first openings **2021** and **3021** are aligned in the vertical direction to form the air conditioning inlet **101**, and the second openings **2022** and **3022** are aligned in the vertical direction to form the air conditioning outlet **102**. The inlet-side air conditioning damper **61** is mounted on the air conditioning inlet **101**, and the outlet-side air conditioning damper **62** is mounted on the air conditioning outlet **102**.

(50) In addition, the air conditioning damper includes damper frames **613** and **623**, damping plates **612** and **622**, and damping motors **611** and **621**.

(51) The damping plates **612** and **622** are rotatably connected to the damper frames **613** and **623**, and the driving shafts of the damping motors **611** and **621** are connected to the rotational shafts of the damping plates **612** and **622**.

(52) FIG. 7 is an exploded perspective view showing the internal configuration of the ventilation apparatus according to an embodiment of the present disclosure.

(53) Referring to FIG. 7, a separate guide duct is not installed in the outdoor air inlet space **S1**. This is because, when the outdoor air introduced into the outdoor air inlet space **S1** passes through the suction surface of the ventilation module **23**, that is, the surface on which the pre-filter **233** is mounted, it is necessary to pass evenly over the entire suction surface. That is, it is more advantageous not to dispose a separate guide duct in order to allow the introduced outdoor air to pass uniformly over the entire suction surface of the ventilation module **23**.

(54) The outdoor air discharge guide duct **25** includes a duct body **251**, a suction fan mounting hole **252** formed on one side surface of the duct body **251**, an air conditioning inlet **254** formed in the bottom surface of the duct body **251**, and an outdoor air outlet **253** formed in the upper surface of the duct body **251**.

(55) The outlet of the suction fan module **21** is connected to the suction fan mounting hole **252**, so that the outdoor air passing through the ventilation module **23** is supplied to the indoor space through the outdoor air outlet **253**.

(56) The lower end of the outdoor air discharge guide duct **25** communicates with the air conditioning outlet **102**, so that the indoor air heat-exchanged while passing through the evaporator **32** is supplied to the indoor space through the outdoor air outlet **253**.

(57) In addition, the indoor air discharge guide duct **26** includes a duct body **261**, an exhaust fan mounting hole **262** formed in one side surface of the duct body **261**, and an indoor air outlet **263** formed in the upper surface of the duct body **261**.

(58) The bottom surface of the duct body **261** has a structure that is not opened. Alternatively, even if opened, the bottom surface of the duct body **261** is shielded by the bottom surface **202** when seated on the lower surface **202** of the upper housing **201**.

(59) The outlet of the exhaust fan module **22** is connected to the exhaust fan mounting hole **262**, so that the indoor air passing through the ventilation module **23** is discharged to the outside through the indoor air outlet **263**.

(60) FIG. **8** is a cutaway perspective view of the indoor air inlet guide duct and the bypass duct taken along line **8-8** of FIG. **7**.

(61) Referring to FIG. **8**, the inlet of the bypass duct **27** is connected to the indoor air inlet guide duct **24**.

(62) In detail, the indoor air inlet guide duct **24** includes a duct body **241**, a total heat exchange outlet **243** formed on one side surface of the duct body **241**, an air conditioning outlet **244** formed at the lower end of the duct body **241**, an indoor air inlet **242** formed at the upper end of the duct body **241**, and a bypass hole **246** formed on the other side surface of the duct body **241**.

(63) A total heat exchange damper module **245** is mounted on the total heat exchange outlet **243**, and a bypass damper module **247** is mounted on the bypass hole **246**.

(64) The total heat exchange damper module **245** and the bypass damper module **247** selectively operate according to the operation mode to open or close the total heat exchange outlet **243** and the bypass hole **246**. The air conditioning outlet **244** communicates with the air conditioning inlet **101**.

(65) With this structure, the indoor air introduced into the indoor air inlet **242** passes through at least one of the total heat exchange outlet **243**, the air conditioning outlet **244**, and the bypass hole **246** according to the operation mode.

(66) FIG. **9** is a side view of the guide duct showing the installation structure of the guide duct and the duct flange, FIG. **10** is a plan view of the guide duct showing the installation structure of the guide duct and the duct flange, and FIG. **11** is a plan view of the duct flange from which the duct flange is removed.

(67) Referring to FIGS. **9** to **11**, the guide duct according to an embodiment of the present disclosure is designed in a shape in which the cross-sectional area increases from the bottom to the top. Hereinafter, the outdoor air discharge guide duct **25** among the three guide ducts will be described as an example. That is, although the outdoor air discharge guide duct **25** will be described as a representative, the same applies to the indoor air inlet guide duct **24** and the indoor air discharge guide duct **26**.

(68) In detail, the duct flange **50** includes a seating portion **501** having a circular ring shape, and a duct connection portion **502** extending upward from the inner edge of the seating portion **501**. The seating portion **501** is in close contact with the upper surface of the upper housing **201**, and the connection portion **502** extends upward from the inner edge of the seating portion **501**.

(69) The seating portion **501** is mounted on the flange connector **2010** formed in the upper housing **201**. The diameter of the flange connector **2010** may be designed to have a length corresponding to the inner diameter of the seating portion **501**, or may be designed to be larger than the inner diameter of the seating portion **501**.

(70) The outdoor air discharge guide duct **25** includes a straight pipe portion extending in a hollow rectangular parallelepiped shape from the lower end to a predetermined height, and an expanded pipe portion in which the cross-sectional area is gradually increased at the upper end of the straight pipe portion. To this end, one or more of the four surfaces of the outdoor air discharge guide duct **25** extends to be rounded with a predetermined curvature, so that the cross-sectional area of the upper end of the outdoor air discharge guide duct **25** is designed to be larger than the cross-sectional area of the lower end.

(71) A line **L1** passing through the center of the outlet of the suction fan module **21** crosses a line **L2** passing through the center of the duct flange **50** (here, the outdoor air outlet flange **52**). In other words, the outlet of the suction fan module **21** is coupled to the outdoor air discharge guide duct **25**

to face the side surface of the upper housing **201**, and the outlet of the exhaust fan module **22** is also coupled to the indoor air discharge guide duct **26** to face the side surface of the upper housing **301**.

(72) In the case of the conventional ventilation apparatus, the center of the flange connector **2010** and the center of the outlet of the fan module are aligned on the same line. At this time, when the shape and size of the outlet of the fan module are designed to be different from the shape and size of the flange connector **2010**, flow resistance occurs inside the duct flange **50**, causing a phenomenon in which air is not smoothly discharged.

(73) However, in the ventilation unit **20** according to the present disclosure, since the outlet of the fan module and the outlet of the duct flange are not aligned on the same line and are directed in directions crossing each other. Therefore, the flow resistance problem that occurs in the conventional ventilation apparatus does not occur.

(74) In addition, since the area of the upper end of the guide duct to which the duct flange is mounted is designed to be significantly larger than the diameter of the duct flange, there is an advantage in that the design freedom of the duct flange is increased.

(75) Meanwhile, the duct flange **50** may be mounted adjacent to the edge of the outdoor air discharge guide duct **25**, and a round portion **255** may be formed at the corner of the outdoor air discharge guide duct **25** on the side on which the duct flange is seated. The round portion **255** may be formed to be rounded with the same radius of curvature as the radius of the inner edge **5011** of the seating portion **501**.

(76) The center of the flange connector **2010** may be designed to be aligned on the same vertical line as the center of the seating portion **501**, and the diameter of the flange connector **2010** and the inner diameter of the seating portion **501** may be designed to have the same size.

(77) The corner of the guide duct on which the round portion **255** is formed may be the corner adjacent to the center of the ventilation unit **20**. As a result, the four duct flanges can be mounted at positions as close as possible to the center of the upper surface of the ventilation unit **20**.

(78) When the guide duct is a duct for guiding the discharge of outdoor air or indoor air, the duct flange is located at the end of the outlet of the guide duct. Accordingly, flow resistance can be minimized.

(79) For example, when the outdoor air discharge flange **52** is mounted on the outer edge b of the outdoor air outlet **253**, a vortex is generated in the space between the outdoor air discharge flange **52** and the inner edge a of the outdoor air outlet **253**, thereby increasing the flow resistance.

(80) Hereinafter, the flow of indoor air and outdoor air made for each operation mode in the ventilation apparatus according to the embodiment of the present disclosure will be described with reference to the drawings.

(81) FIG. **12** is a view showing air flow inside the ventilation apparatus in a total heat exchange ventilation mode.

(82) Referring to FIG. **12**, when the total heat exchange ventilation mode is executed, both the suction fan module **21** and the exhaust fan module **22** operate and the total heat exchange damper module **245** operates, and thus, the total heat exchange outlet **243** of the indoor air inlet guide duct **24** is opened. The air conditioning inlet **101** and the air conditioning outlet **102** are maintained in a closed state, and the evaporation fan module **34** is not driven.

(83) In this state, when the suction fan module **21** is driven, outdoor air is introduced into the outdoor air inlet space **S1** through the outdoor air inlet duct **41**. The outdoor air introduced into the outdoor air inlet space **S1** passes through the ventilation module **23** and is then guided to the outdoor air discharge space **S2**. The outdoor air guided to the outdoor air discharge space **S2** is sucked by the suction fan module **21** and is supplied to the indoor space through the outdoor air discharge guide duct **25** and the outdoor air discharge duct **42**.

(84) When the exhaust fan module **22** is driven, the indoor air is guided to the indoor air inlet guide duct **24** through the indoor air inlet duct **43**, and is discharged to the indoor air inlet space **S3**

through the total heat exchange outlet **243**. The indoor air discharged to the indoor air inlet space **S3** passes through the total heat exchange element **231** and is then guided to the indoor air discharge space **S4**. The outdoor air and the indoor air exchange heat while passing through the ventilation module **23** (waste heat recovery).

(85) The indoor air guided to the indoor air discharge space **S4** is sucked by the exhaust fan module **22** and then discharged to the outside through the indoor air discharge guide duct **26** and the indoor air discharge duct **44**.

(86) In winter, by the waste heat recovery process occurring in the ventilation module **23**, the outdoor air absorbs heat from the indoor air discharged to the outside and is then supplied to the indoor space, thereby preventing a sudden drop in the indoor temperature.

(87) On the other hand, in summer, heat is emitted from the outdoor air to the indoor air discharged to the outside and supplied to the indoor space in a low temperature state, thereby preventing a sudden increase in the indoor temperature.

(88) FIG. **13** is a view showing air flow inside the ventilation apparatus in a quick ventilation mode.

(89) Referring to FIG. **13**, when the quick ventilation mode (or outdoor air cooling mode) is executed, the suction fan module **21** and the exhaust fan module **22** operate, and the evaporation fan module **34** does not operate.

(90) The total heat exchange outlet **243**, the air conditioning inlet **101**, and the air conditioning outlet **102** are closed, and the bypass damper module **247** operates to open the bypass hole **246**.

(91) When the suction fan module **21** operates in this state, outdoor air sequentially passes through the outdoor air inlet duct **41**, the outdoor air inlet space **S1**, the ventilation module **23**, the outdoor air discharge space **S2**, the suction fan module **21**, the outdoor air discharge guide duct **25**, and the outdoor air discharge and is supplied to the indoor space.

(92) When the exhaust fan module **22** operates, the indoor air sequentially passes through the indoor air inlet duct **43**, the indoor air inlet guide duct **24**, the bypass duct **27**, the indoor air discharge space **S4**, the exhaust fan module **22**, the indoor air discharge guide duct **26**, and the indoor air discharge duct **44** and is discharged to the outside.

(93) In this case, since the indoor air does not pass through the ventilation module **23**, the outdoor air introduced into the indoor space is discharged to the indoor space at an outdoor temperature. Accordingly, when the quick ventilation mode is executed in autumn or winter, fresh outdoor air having a low temperature is supplied to the indoor space.

(94) In addition, since the indoor air does not pass through the ventilation module **23**, oil, smoke, and other harmful substances scattered in the air during the cooking process are directly discharged to the outside during cooking a kitchen, thereby minimizing contamination of the ventilation module **23**.

(95) When the total heat exchange ventilation mode is executed during cooling in the kitchen, the inner circumferential surface of the total heat exchange element of the ventilation module **23** is contaminated while the contaminated indoor air passes through the ventilation module **23**. As a result, the life of the total heat exchange element may be shortened.

(96) Accordingly, when the indoor air is heavily contaminated, the quick ventilation mode using the bypass duct is executed to allow the indoor air to be quickly discharged to the outside and to minimize contamination of the total heat exchange element.

(97) FIG. **14** is a view showing air flow inside the ventilation apparatus in a quick cooling mode.

(98) Referring to FIG. **14**, when the quick cooling mode is executed, the suction fan module **21** and the exhaust fan module **22** are stopped and only the evaporation fan module **34** operates.

(99) The inlet-side air conditioning damper **61** and the outlet-side air conditioning damper **62** operate to open both the air conditioning inlet **101** and the air conditioning outlet **102**.

(100) The total heat exchange outlet **243** and the bypass hole **246** maintain a closed state.

(101) When the evaporation fan module **34** operates in this state, the indoor air passes through the

indoor air inlet duct **43**, the indoor air inlet guide duct **24**, and the air conditioning inlet **101**, and is guided to the inside of the air conditioning unit **30** in which the evaporation fan module **34** is installed.

(102) The indoor air guided to the air conditioning unit passes through the air conditioning outlet **102** in a state in which the temperature thereof is lowered while passing through the evaporation fan module **34**, and is discharged again to the indoor space through the outdoor air outlet guide duct **25** and the outdoor air discharge duct **42**.

(103) In addition, a damper module may be mounted in the suction fan mounting hole **252** of the outdoor air discharge guide duct **25**. In the quick cooling mode, the suction fan mounting hole **252** may be maintained in a closed state by the damper module.

(104) FIG. **15** is a view showing air flow inside the ventilation apparatus in a total heat exchange ventilation/cooling simultaneous mode.

(105) Referring to FIG. **15**, when the total heat exchange ventilation/cooling simultaneous mode is executed, the total heat exchange ventilation mode and the quick cooling mode are simultaneously performed.

(106) To this end, the suction fan module **21**, the exhaust fan module **22**, and the evaporation fan module **34** all operate, the total heat exchange outlet **243**, the air conditioning inlet **101**, and the air conditioning outlet **102** are opened, and the bypass hole **246** is closed.

(107) In this state, the exhaust fan module **22** operates so that the indoor air flowing along the indoor air inlet duct **43** and the indoor air inlet guide duct **24** flows into the total heat exchange outlet **243** and the air conditioning inlet **101**. The indoor air discharged through the total heat exchange outlet **243** passes through the total heat exchange element **231** and is discharged to the outside, and the indoor air discharged through the air conditioning inlet **101** is supplied back into the indoor space through the evaporator **34**, the outdoor air discharge guide duct **25**, and the outdoor air discharge duct **42**.

(108) In addition, the suction fan module **21** operates so that outdoor air is guided to the outdoor air inlet space **S1** through the outdoor air inlet duct **41** and is guided to the ventilation module **23** and the outdoor air discharge space **S2**. The outdoor air is guided to the outdoor air discharge guide duct **25** by the suction fan module **21**, is mixed with low-temperature indoor air passing through the evaporator **34** in the outdoor air discharge guide duct **25**, and is supplied to the indoor space through the outdoor air discharge duct **42**.

(109) FIG. **16** is a view showing air flow inside the ventilation apparatus in a quick ventilation/cooling simultaneously mode.

(110) Referring to FIG. **16**, when the quick ventilation/cooling simultaneous mode is executed, the quick ventilation mode and the cooling mode are simultaneously performed.

(111) To this end, the suction fan module **21**, the exhaust fan module **22**, and the evaporation fan module **34** all operate, and the total heat exchange outlet **243** is closed. On the other hand, the air conditioning inlet **101**, the air conditioning outlet **102**, and the bypass hole **246** are opened.

(112) In this state, the indoor air introduced into the indoor air inlet duct **43** when the suction fan module **21** operates flows into the air conditioning inlet **101** and the bypass hole **246** inside the indoor air inlet guide duct **24**.

(113) The indoor air guided to the bypass duct **27** through the bypass hole **246** is discharged to the outside through the indoor air discharge space **S4**, the indoor air discharge guide duct **26**, and the indoor air discharge duct **44**.

(114) On the other hand, the indoor air passing through the air conditioning inlet **101** passes through the evaporation fan module **34** and the evaporator **32** and is supplied back to the indoor space through the outdoor air guide duct **25** and the outdoor air discharge duct **42**.

(115) The exhaust fan module **22** operates so that outdoor air is guided to the outdoor air inlet duct **41**, the outdoor air inlet space **S1**, the ventilation module **23**, and the outdoor air discharge space **S2**. The outdoor air guided to the outdoor air discharge space **S2** is guided to the outdoor air

discharge guide duct **25**, is mixed with the indoor air passing through the evaporator **32**, and is supplied to the indoor space through the outdoor air discharge duct **42**.

(116) Meanwhile, a compressor, a condenser, and a four-way valve constituting the refrigerant cycle are disposed outside the ventilation apparatus **10**. Therefore, there is an advantage in that the load of the ventilation apparatus **10** is relatively light and the cooling mode and the heating mode can be selectively driven through the opening/closing control of the four-way valve.

(117) In other words, when the refrigerant cycle operates as a heat pump through the opening/closing control of the four-way valve, the evaporator **32** functions as a condenser, and thus, there is an advantage in that indoor heating is also possible.

(118) FIG. **17** is an exploded perspective view of the ventilation apparatus showing a state in which the ventilation unit and the air conditioning unit are connected by a connection unit, and FIG. **18** is a longitudinal cross-sectional view taken along line **18-18** of FIG. **1**.

(119) Referring to FIGS. **17** and **18**, the ventilation unit **20** and the air conditioning unit **30** are detachably connected by the connection unit **70**.

(120) In detail, the connection unit **70** includes a ventilation-side flange **71** coupled to two opposite side surfaces of the upper housing **201** constituting the outer shape of the ventilation unit **20**, an air conditioning-side flange **72** coupled to two opposite sides of the lower housing **301** constituting the outer shape of the air conditioning unit **30**, and a coupler **73** for connecting the ventilation-side flange **71** and the air conditioning-side flange **72**.

(121) The connection unit **70** may be formed to have a length equal to or shorter than the length from the front to the rear of the ventilation apparatus **10**.

(122) The side surface of the upper housing **201** to which the ventilation-side flange **71** is coupled and the side surface of the lower housing **301** to which the air conditioning-side flange **72** is coupled pass through the same vertical plane.

(123) The ventilation-side flange **71** is coupled to the lower end of the side surface of the upper housing **201**, and the air conditioning-side flange **72** is coupled to the upper end of the side surface of the lower housing **201**.

(124) The longitudinal cross-sections of the ventilation-side flange **71** and the air conditioning-side flange **72** may be symmetrical with respect to a horizontal plane.

(125) Specifically, the ventilation-side flange **71** includes a close contact portion **711** in close contact with the side surface of the ventilation apparatus **10**, a bent portion **712** extending horizontally from the lower end of the contact portion **711** in a direction away from the side surface of the upper housing **201**, and an extension portion **713** bent upward from the end of the bent portion **712**.

(126) In addition, the air conditioning-side flange **72** has a cross section symmetrical to the ventilation-side flange **71**, and includes a close contact portion **721**, a bent portion **722**, and an extension portion **723**.

(127) The coupler **73** may have a cross-sectional shape that is bent multiple times to surround the two extension portions **723**. That is, the coupler **73** may be bent twice by 90 degrees at two positions spaced apart by the same distance from the center of the short side, and may have a cross-sectional shape in which the two long sides are spaced apart. The two long sides may be spaced apart by a distance corresponding to the sum of the thicknesses of the bent portions **712** and **722**.

(128) In order to couple the ventilation unit **20** and the air conditioning unit **30**, the ventilation unit **20** is stacked on the upper surface of the air conditioning unit **30**. Accordingly, the bent portion **712** of the ventilation-side flange **71** and the bent portion **722** of the air conditioning-side flange **72** are in close contact with each other, and the extension portion **713** of the ventilation-side flange **71** and the extension portion **723** of the air conditioning-side flange **72** extend in opposite directions.

(129) In this state, the rear end of the coupler **73** is positioned at the front end of the flanges **71** and **72** and then the coupler **73** is moved in the longitudinal direction of the flanges **71** and **72**.

(130) In other words, the bent portions **712** and **722** are fitted in the space formed between the two

long sides of the coupler **73**, and the flanges **71** and **72** and the coupler **73** are aligned on the same horizontal line so that the extension portions **713** and **723** are in a state that can be accommodated in the coupler **73**. In this state, when the rear end of the coupler **73** is slid backward, the extension portions **713** and **723** are completely inserted into the coupler **73**. Accordingly, the ventilation-side flange **71** and the air conditioning-side flange **72** are not separated and maintain a combined state. (131) The coupler **73** may be made of a metal plate having a strength that does not spread by the load of the ventilation unit **30**.

Claims

1. A ventilation apparatus comprising: a housing defining an inner space, the housing having an outdoor air inlet, an outdoor air outlet, an indoor air inlet, and an indoor air outlet; a partition configured to divide the inner space of the housing into a ventilation space and an air conditioning space; a total heat exchange element located in the ventilation space, the total heat exchange element having a hexahedral shape with four side edges respectively disposed to face four side surfaces of the housing; four partition walls configured to respectively connect the four side edges of the total heat exchange element and the four side surfaces of the housing to partition the ventilation space, except for a space for the total heat exchange element, into an outdoor air inlet space, an outdoor air discharge space, an indoor air inlet space, and an indoor air discharge space; an outdoor air discharge guide duct disposed in the outdoor air discharge space; an indoor air inlet guide duct disposed in the indoor air inlet space; an indoor air discharge guide duct disposed in the indoor air discharge space; a suction fan disposed in the outdoor air discharge space, the suction fan being configured to draw outdoor air through the outdoor air inlet; an exhaust fan disposed in the indoor air discharge space, the exhaust fan being configured to draw indoor air through the indoor air inlet; a bypass duct branching from the indoor air inlet guide duct, the bypass duct being configured to bypass the total heat exchange element, the bypass duct extending to the indoor air discharge space; a boundary wall configured to partition the air conditioning space into a first space and a second space; a heat exchanger located in the first space; and an air conditioning fan disposed in the second space.
2. The ventilation apparatus of claim 1, wherein the bypass duct is branched from a first side surface of the indoor air inlet guide duct.
3. The ventilation apparatus of claim 2, wherein a total heat exchange outlet is formed on a second side surface of the indoor air inlet guide duct, and wherein the ventilation apparatus further comprises a total heat exchange damper located at the total heat exchange outlet, the total heat exchange damper being configured to selectively open and close the total heat exchange outlet such that the indoor air inlet guide duct and the indoor air inlet space selectively communicate with each other.
4. The ventilation apparatus of claim 2, wherein the bypass duct is bent along an upper surface of the partition and an inner edge of the housing, the bypass duct being in communication with the indoor air outlet.
5. The ventilation apparatus of claim 4, wherein the bypass duct includes an inlet, and wherein the ventilation apparatus further comprises a bypass damper located at the inlet of the bypass duct.
6. The ventilation apparatus of claim 2, wherein the outdoor air inlet is in communication with the outdoor air inlet space, wherein the outdoor air outlet is in communication with an upper surface of the outdoor air discharge guide duct, wherein the indoor air inlet is in communication with an upper surface of the indoor air inlet guide duct, and wherein the indoor air outlet is in communication with an upper surface of the indoor air discharge guide duct.
7. The ventilation apparatus of claim 1, further comprising: a plurality of duct flanges on an upper surface of the housing; and a plurality of air ducts connected to the plurality of duct flanges, each of the air ducts extending from the housing.

8. The ventilation apparatus of claim 7, wherein the plurality of duct flanges comprises: an outdoor air inlet flange located at the outdoor air inlet; an outdoor air discharge flange located at the outdoor air outlet; an indoor air inlet flange located at the indoor air inlet; and an indoor air discharge flange located at the indoor air outlet.
9. The ventilation apparatus of claim 8, wherein the plurality of air ducts comprises: an outdoor air inlet duct connected to the outdoor air inlet flange; an outdoor air discharge duct connected to the outdoor air discharge flange; an indoor air inlet duct connected to the indoor air inlet flange; and an indoor air discharge duct connected to the indoor air discharge flange.
10. The ventilation apparatus of claim 1, wherein the partition includes: an air conditioning inlet configured to connect the indoor air inlet guide duct and the first space; and an air conditioning outlet configured to connect the outdoor air discharge guide duct and the second space, and wherein the ventilation apparatus further comprises: an air conditioning inlet damper located at the air conditioning inlet; and an air conditioning outlet damper located at the air conditioning outlet.
11. The ventilation apparatus of claim 1, wherein a total heat exchange outlet is formed on a side surface of the indoor air inlet guide duct, and wherein the ventilation apparatus further comprises a total heat exchange damper located at the total heat exchange outlet, the total heat exchange damper being configured to selectively open and close the total heat exchange outlet such that the indoor air inlet guide duct and the indoor air inlet space selectively communicate with each other.
12. The ventilation apparatus of claim 1, further comprising at least one of a HEPA filter and a pre-filter located between a side surface of the total heat exchange element and the outdoor air inlet space.
13. The ventilation apparatus of claim 1, wherein the housing includes: an upper housing having a lower surface; and a lower housing having an upper surface, wherein the lower surface of the upper housing and the upper surface of the lower housing define the partition dividing the inner space of the housing.
14. The ventilation apparatus of claim 13, further comprising a connection unit to connect the upper housing to the lower housing at outer surfaces thereof.
15. A ventilation apparatus comprising: a housing defining an inner space, the housing having an outdoor air inlet, an outdoor air outlet, an indoor air inlet, and an indoor air outlet; a plurality of air ducts extending from the housing; a partition configured to divide the inner space of the housing into a ventilation space and an air conditioning space, the partition having an air conditioning inlet and an air conditioning outlet; a total heat exchange element located in the ventilation space, the total heat exchange element having four side edges respectively disposed to face four side surfaces of the housing; four partition walls configured to respectively connect the four side edges of the total heat exchange element and the four side surfaces of the housing to partition the ventilation space, except for a space for the total heat exchange element, into an outdoor air inlet space, an outdoor air discharge space, an indoor air inlet space, and an indoor air discharge space; an outdoor air discharge guide duct disposed in the outdoor air discharge space; an indoor air inlet guide duct disposed in the indoor air inlet space; an indoor air discharge guide duct disposed in the indoor air discharge space; a suction fan disposed in the outdoor air discharge space, the suction fan being configured to draw outdoor air through the outdoor air inlet; an exhaust fan disposed in the indoor air discharge space, the exhaust fan being configured to draw indoor air through the indoor air inlet; a bypass duct branching from indoor air inlet guide duct, the bypass duct being configured to bypass the total heat exchange element, the bypass duct extending to the indoor air discharge space; a heat exchanger located in the air conditioning space; and an air conditioning fan located in the air conditioning space.
16. The ventilation apparatus of claim 15, wherein the plurality of air ducts comprises: an outdoor air inlet duct connected to the outdoor air inlet space; an outdoor air discharge duct connected to the outdoor air discharge space; an indoor air inlet duct connected to the indoor air inlet space; and an indoor air discharge duct connected to the indoor air discharge space.

17. The ventilation apparatus of claim 16, wherein the bypass duct is branched from a first side surface of the indoor air inlet guide duct.

18. The ventilation apparatus of claim 15, further comprising at least one of a HEPA filter and a pre-filter located between a side surface of the total heat exchange element and the outdoor air inlet space.

19. The ventilation apparatus of claim 15, wherein the housing includes: an upper housing having a lower surface; and a lower housing having an upper surface, wherein the lower surface of the upper housing and the upper surface of the lower housing define the partition dividing the inner space of the housing.
